The Informational Diet of /pol/

A citation-level audit of source use and discourse tone
across 278,706 posts on 4chan's Politically Incorrect board

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Abstract

Research on 4chan's /pol/ board routinely frames its information environment as a contest between mainstream and alternative news media. We test that framing empirically. Over 24 daily captures spanning 45 days (2026-07-22 to 2026-09-04) we collected 6,532 threads and 278,706 posts, extracted every external URL, and coded 10,078 citation events across 1,408 unique domains, attaching a transformer sentiment score to the discussion surrounding each citation.

The framing does not survive contact with the data. News outlets of any kind—mainstream, alternative, and state-controlled combined—account for **4.6%** of citation events. What /pol/ actually circulates is platform content (40.3%) and a long, largely unclassifiable tail (47.2%), of which file hosts and archives are a substantial and behaviourally meaningful component (9.2% of all citations). Sourcing is extremely concentrated: a Gini coefficient of 0.821, with YouTube alone supplying 28.9% of citations and twelve domains supplying half.

Discourse tone differs across source categories (Kruskal–Wallis $H = 110.0$, $p < 10^{-21}$), but the effect is substantively small ($\varepsilon^2 = 0.011$; adjusted $R^2 = 0.067$). Mainstream citations sit in measurably more negative discussion than platform, unclassified, or institutional citations (Cliff's δ between 0.21 and 0.26, all $p_{\text{Holm}} < 10^{-7}$). The specific mainstream-versus-alternative contrast that motivates the literature, however, is **not supported**: $p_{\text{Holm}} = 0.83$, $\delta = -0.121$ (negligible), with $n = 69$ alternative citations affording essentially no power.

We also convert the design's primary validity threat into a measurement. Across 62 pairs of repeated thread captures, 0.68% of posts (95% CI [0.44, 1.07]) disappeared between observations—a bound that is itself conditioned on thread survival and therefore a floor, not an estimate. We argue that the mainstream/alternative frame is the wrong instrument for this object, and that platform and archive infrastructure is the more defensible unit of analysis.

1 Introduction

4chan's /pol/ ("Politically Incorrect") board occupies an outsized place in accounts of online radicalisation and information disorder. Measurement work has characterised its language, its raiding behaviour, and its position in the wider web ecosystem (Hine et al., 2017; Papasavva et al., 2020). A recurring assumption in this literature, and in journalistic accounts drawing on it, is that /pol/ constitutes an *alternative media ecosystem*: a space where legacy outlets are rejected and replaced by a counter-canon of partisan publishers.

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That assumption is rarely measured directly. It requires knowing what /pol/ cites, in what proportion, and how the surrounding discussion treats each kind of source. This paper supplies that measurement for a bounded collection window and reports a result that is largely negative for the framing.

Our contributions are four:

1. **A citation-level audit.** 10,078 citation events over 45 days, coded by medium type, with proportions reported as Wilson score intervals.

2. **A concentration analysis.** Gini, normalised HHI, and a Lorenz curve over the domain distribution, quantifying how narrow the diet is.

3. **An inferential test of the tone hypothesis** with effect sizes and a thread-clustered adjusted model—including an explicit null on the mainstream/alternative contrast.

4. **A measured bound on survivorship bias**, derived from repeated captures of the same threads, in place of the usual qualitative caveat.

2 Related work

Hine et al. (2017) provided the first large-scale measurement of /pol/, characterising its posting dynamics, its hate-speech prevalence, and its raiding of other platforms. Papasavva et al. (2020) released a 3.5-year corpus of augmented /pol/ posts, establishing the dataset conventions much subsequent work relies on. Both establish /pol/ as an object of quantitative study; neither takes the citation event as the unit of analysis.

Source-classification work in computational social science generally anchors labels to third-party rating databases rather than author judgment—AllSides, Media Bias/Fact Check, and Ad Fontes are the common choices (Bozarth et al., 2020; Robertson et al., 2018). We adopt that principle, with the substantial caveat noted in Section 6: our present implementation codes only the medium-type axis, and the ratings-anchored orientation and ownership axes remain specified but unimplemented.

For sentiment we use the TweetEval RoBERTa model (Barbieri et al., 2020; Loureiro et al., 2022). Its limitations on this domain are substantial and we treat them as a first-order threat rather than a footnote (Section 6).

3 Data and methods

3.1 Collection

Threads were captured from the live /pol/ JSON API via the 4TCT collector on a daily cadence, yielding 24 daily captures between 2026-07-22 and 2026-09-04. This is a *snapshot* design: conclusions are scoped to the collection window, not to /pol/ in general or across time.

3.2 Citation extraction and unit of analysis

The unit of analysis is the **citation event**: one external domain appearing in one post. Posts may contain several distinct domains; each is counted once per post, deduplicated within the post. Links internal to 4chan are excluded. Mobile and shortened hosts are normalised to a canonical domain (`youtu.be`, `m.youtube.com` → `youtube.com`), and subdomain matching resolves cases such as `en.wikipedia.org`.

Threads captured	6,532
Posts	278,706
Posts with ≥ 1 external link	7,107 (2.55%)
Citation events	10,078
Unique domains	1,408
Daily captures	24
Window	2026-07-22 – 2026-09-04

Table 1: Corpus summary.

We note that an earlier version of our pipeline retained only the first domain per post and discarded the remainder. Because /pol/ posts characteristically lead with a video or tweet and append a source, this systematically discarded the appended source: alternative-media citations were undercounted by a factor of 3.6 (19 recorded against 69 actual). All figures reported here use per-citation counting, and we flag the failure mode because it is a plausible silent defect in any pipeline that treats a post as carrying one source.

3.3 Source coding

Domains are coded on the *medium type* axis by deterministic rules: institutional/reference (Wikimedia family; `.gov`, `.mil`, `.edu` by suffix); social platform (curated platform set); news outlet, subdivided into mainstream, alternative, and state-controlled; and unclassified. Section 6 states plainly what this coding does and does not license.

3.4 Sentiment measurement

Each post carrying at least one citation is scored with `cardiffnlp/twitter-roberta-base-sentiment-latest` (Barbieri et al., 2020; Loureiro et al., 2022). The three-way softmax is collapsed to a compound score in $[-1, +1]$ as $s = p_{\text{pos}} - p_{\text{neg}}$. A post is scored once; the score is attached to each of its citation events.

Critically, the score is computed over the *entire post*. It measures the **ambient tone of discussion surrounding a citation**, not the author's stance toward the cited source. "CNN is garbage, here's proof" yields a negative score attached to CNN even though the poster is endorsing the link's content. Every tone result below must be read under that construct definition.

3.5 Statistical analysis

Proportions are reported with 95% Wilson score intervals (Wilson, 1927), which behave correctly for the small and extreme cells this distribution produces. Concentration is measured by the Gini coefficient and a normalised Herfindahl–Hirschman index over per-domain citation counts.

For tone we use Kruskal–Wallis (Kruskal & Wallis, 1952) as the omnibus test—the sentiment distribution is bounded and non-normal—with ε^2 as the effect size, followed by Dunn post-hoc tests with tie correction and Holm step-down adjustment (Holm, 1979). Pairwise effect sizes are Cliff's δ (Cliff, 1993), interpreted at the conventional thresholds (negligible < 0.147, small < 0.33, medium < 0.474). We report effect sizes throughout: at $n = 10{,}078$, statistical significance is cheap and carries little substantive information on its own.

The adjusted model is an OLS regression of compound sentiment on source category, thread topic, and $\log(1 + \text{post length})$, with standard errors clustered by thread, since citations within a thread are not independent.

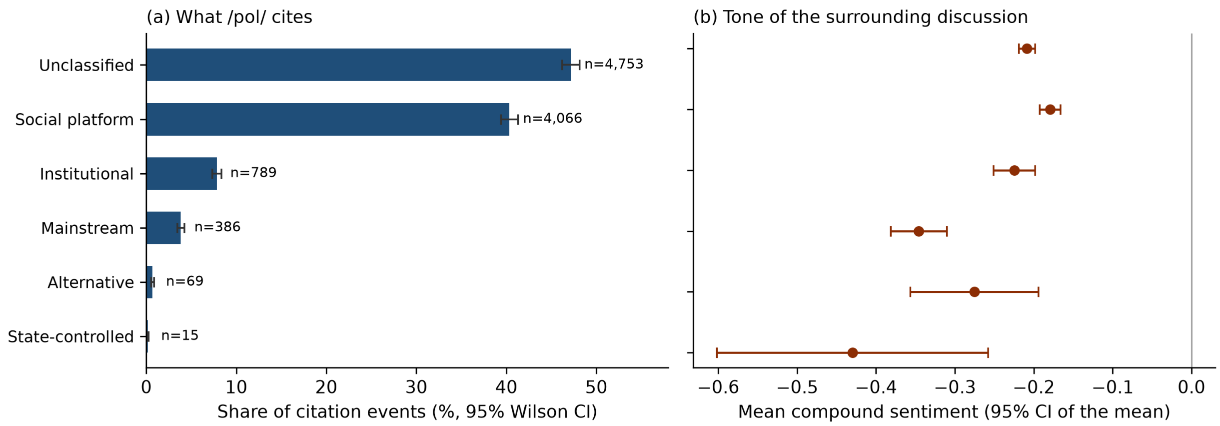

Figure 1: (a) Share of citation events by source category, with 95% Wilson score intervals. (b) Mean compound sentiment of the surrounding discussion, with 95% confidence intervals *of the mean*. The width of the Alternative and State-controlled intervals in (b) is the whole story of the null result in Section 4.4.

Category	n	Share	95% CI	Mean sent.	Neg. ratio
Unclassified	4,753	47.2%	[46.2, 48.1]	−0.209	46.5%
Social platform	4,066	40.3%	[39.4, 41.3]	−0.179	39.4%
Institutional	789	7.8%	[7.3, 8.4]	−0.225	45.2%
Mainstream	386	3.8%	[3.5, 4.2]	−0.346	66.1%
Alternative	69	0.7%	[0.5, 0.9]	−0.275	62.3%
State-controlled	15	0.1%	[0.1, 0.2]	−0.430	86.7%

Table 2: Citation events by source category. "Neg. ratio" is the share of citations whose surrounding post scores below −0.2.

4 Results

4.1 What /pol/ cites

Figure 1(a) and Table 2 give the distribution. The result that organises everything else: **news outlets of any description account for 4.6% of citation events**. Mainstream outlets supply 3.8% [3.5, 4.2], alternative outlets 0.7% [0.5, 0.9], and state-controlled outlets 0.1% [0.1, 0.2]. Social platforms supply 40.3%, and 47.2% of citations fall outside any of these categories.

These proportions are stable across the window. Daily mainstream share ranges from 1.8% to 6.0% (median 3.7%) and daily platform share from 33.5% to 51.5% (median 40.4%) across the 24 captures, with no visible trend.

4.2 Concentration

The diet is not merely non-journalistic; it is narrow. The domain distribution has a Gini coefficient of **0.821** and a normalised HHI of 0.091. YouTube alone supplies **28.9%** of all citation events; the top ten domains supply 49.1% and the top fifty 69.0%. Twelve domains account for half of all citations. At the other end, 989 of 1,408 domains (70.2%) appear exactly once. Figure 2 plots the Lorenz curve.

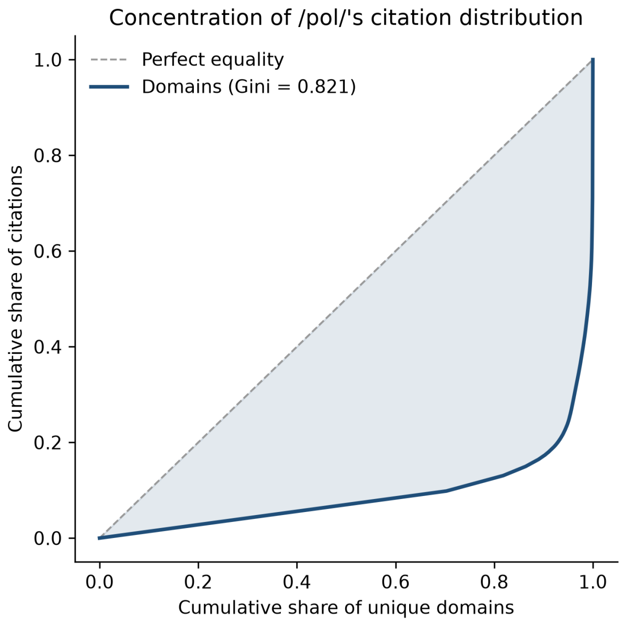

Figure 2: Lorenz curve of the citation distribution over 1,408 unique domains.

4.3 Archives and file hosts are a sourcing behaviour

The 47.2% "unclassified" bucket is not homogeneous noise. File hosts and archiving services account for **928 citation events**—9.2% of all citations and 19.5% of the unclassified bucket. The leading members are `files.catbox.moe` (253), `rentry.org` (168), `archive.4plebs.org` (117), `archive.today` (83), `archive.ph` (66) and `archive.is` (63).

Read as infrastructure rather than residue, this is a coherent practice: hosting media outside platform moderation, and preserving content—including 4chan's own deleted threads, via 4plebs—against removal. On this board, archive use is roughly *twice* as frequent as citing mainstream news.

4.4 Discourse tone by source category

Tone differs across categories: Kruskal–Wallis $H = 110.0$, $p = 4.0 \times 10^{-22}$. The effect size, however, is $\varepsilon^2 = 0.011$—source category accounts for roughly one percent of the variance in sentiment. The adjusted model reaches $R^2 = 0.067$. Whatever is driving tone on /pol/, the category of the cited source is a minor part of it.

Table 3 gives the post-hoc structure. Mainstream citations sit in reliably more negative discussion than platform ($\delta = 0.255$), unclassified ($\delta = 0.242$), and institutional ($\delta = -0.208$) citations, all at small effect magnitude and all surviving Holm correction comfortably.

The central contrast is null. Mainstream versus alternative—the comparison the alternative-ecosystem framing predicts—yields $z = -1.09$, $p_{\text{Holm}} = 0.827$, and $\delta = -0.121$, a negligible magnitude. The raw means differ in the predicted direction (-0.346 against -0.275), and reporting that difference without a test would have produced a confident and false headline. With $n = 69$ alternative citations, the study has essentially no power to resolve a gap of this size, and we decline to interpret it.

Contrast	z	p_{Holm}	Cliff's δ	Magnitude
Social platform vs. Mainstream	9.24	$< 10^{-15}$	0.255	small
Unclassified vs. Mainstream	7.10	1.8×10^{-11}	0.242	small
Mainstream vs. Institutional	−5.98	2.9×10^{-8}	−0.208	small
Social platform vs. Unclassified	5.44	6.4×10^{-7}	0.070	negligible
Social platform vs. Institutional	3.09	0.022	0.067	negligible
Social platform vs. Alternative	2.88	0.040	0.181	small
Social platform vs. State-controlled	2.84	0.041	0.368	medium
Mainstream vs. Alternative	−1.09	**0.827**	−0.121	negligible

Table 3: Dunn post-hoc contrasts with Holm correction, ordered by adjusted p. All contrasts above the rule are significant at $\alpha = 0.05$; the contrast below it is the one the literature's framing predicts, and it is not. Contrasts omitted for space were all non-significant.

Term	β	SE	p
Intercept	0.033	0.086	0.70
Mainstream	−0.094	0.022	1.6×10^{-5}
State-controlled	−0.218	0.098	0.026
Institutional	−0.020	0.015	0.18
Alternative	−0.008	0.045	0.86
Unclassified	0.006	0.011	0.57
Topic: Elections / Politics	0.239	0.084	0.004
$\log(1 + \text{post length})$	−0.042	0.005	1.2×10^{-14}

Table 4: OLS regression of compound sentiment, standard errors clustered by thread ($n = 10{,}078$ citations in 2,642 threads; $R^2 = 0.067$). Reference category is social platform; reference topic is Economy / Finance. Remaining topic terms omitted for space.

The adjusted model (Table 4) reproduces this pattern. Relative to social platforms, mainstream citations carry -0.094 ($p = 1.6 \times 10^{-5}$) and state-controlled citations -0.218 ($p = 0.026$), while alternative citations are statistically indistinguishable from platform citations ($\beta = -0.008$, $p = 0.86$). Post length is a stronger and more reliable predictor than most category contrasts: each unit of log length carries -0.042 ($p = 1.2 \times 10^{-14}$). Longer posts are angrier posts.

4.5 Co-citation structure

Domains co-occurring within a thread form a weighted graph (edges with weight ≥ 2): 286 nodes, 1,846 edges, density 0.045. Louvain community detection yields six communities at modularity 0.429. The largest (198 domains) is a general-purpose cluster anchored on YouTube, Wikipedia, and mainstream outlets; smaller ones are topical—a war-tracking cluster (`liveuamap`, `marinetraffic`, `vesselfinder`, `oilprice`), a US-politics platform cluster (`twitter`, `truthsocial`, `rumble`), and a German-language cluster. Sourcing ecosystems on /pol/ appear organised by topic and by platform affordance, not by editorial orientation.

4.6 A measured bound on survivorship

Sixty-two pairs of repeated captures of the same thread let us measure directly what is usually asserted: **19 of 2,776 observed posts (0.68%, 95% CI [0.44, 1.07]) had disappeared by the following capture**, affecting 7 of the paired threads.

This number must be read for what it is. It measures post-level deletion *within threads that survived to be captured twice*. It says nothing about threads deleted in their entirety, nor about threads born and died between two daily captures—which, at a daily cadence on a board with /pol/'s velocity, is the larger loss channel. It is therefore a **floor on content loss, not an estimate of it**, and it is a floor derived from a self-selected sample. We report it because a measured floor is more useful than an unmeasured caveat, not because it settles the question.

5 Discussion

The frame does not fit the object. A research literature that asks whether /pol/ prefers alternative to mainstream media is asking about 4.6% of what /pol/ circulates. The board's information diet is overwhelmingly platform-mediated and archive-supported: a YouTube link with commentary, a Telegram forward, a catbox-hosted image, a 4plebs permalink to a deleted thread. Treating this as a *news* ecosystem—counter or otherwise—imports a print-era category that the object does not honour. The more defensible unit is platform and archival infrastructure.

Significance is not magnitude. Category differences in tone are real and, on the well-populated categories, robust. They are also small: one percent of sentiment variance. A study of this size can detect almost anything; reporting p-values without ε^2 and δ would have dressed a marginal effect as a finding. The mainstream/alternative null is the sharper lesson—the difference sits in the predicted direction and would have been reported as confirmation by any analysis that stopped at group means.

What negativity toward mainstream citations does and does not mean. Mainstream citations reliably attract more negative surrounding discourse. Under our construct definition this is ambient tone, not stance. Both readings are consistent with it: hostility to the outlet, and hostility to the *event* that mainstream outlets disproportionately report. We cannot separate them without target-dependent stance detection (Mohammad et al., 2016; Pontiki et al., 2016), and we do not claim to.

6 Limitations

We state these at full strength; several are severe enough to bound what the paper can claim.

Sentiment measurement is unvalidated on this domain. The model was trained on Twitter. /pol/'s irony, slang, and toxicity are out-of-distribution, and we have *not* performed the human-annotated validation that would quantify the resulting measurement error. Every tone result in Section 4.4 should be treated as provisional pending a hand-coded sample of 100–200 posts with reported model–human agreement. The descriptive and concentration results (Sections 4.1–4.3) do not depend on the sentiment model and are unaffected.

Source coding is single-axis and partially unanchored. Our design calls for coding each domain on three orthogonal axes—medium type, editorial orientation, and ownership/control—against external rating databases (AllSides, MBFC, Ad Fontes, RSF), precisely because forcing them into one label is what makes cases like Fox News, BBC, and France 24 irreconcilable. The implementation reported here codes only the medium-type axis, and its news-outlet subdivision into mainstream/alternative/state-controlled is a curated list, not a ratings-anchored mapping. No inter-annotator agreement was computed. The mainstream/alternative/state distinctions therefore carry lower construct validity than the platform, institutional, and archive distinctions, which rest on deterministic structural rules.

Survivorship bias is bounded below, not corrected. See Section 4.6. Moderation on /pol/ removes the most extreme content preferentially, so the surviving corpus is a filtered view—filtered in the direction of the phenomenon being measured. Source-distribution estimates are relatively robust to this, since deletion would need to correlate with the category of the cited source. Tone estimates are directly exposed, because deletion correlates with discourse extremity, which is what the sentiment score captures.

Small cells preclude inference. $n = 69$ (alternative) and $n = 15$ (state-controlled) support description only. No amount of additional collection fixes this efficiently: at 0.7% and 0.1% of citations, reaching usable power would require an order of magnitude more data or a broader ratings-anchored domain list. The latter is the correct fix.

Topic classification is weak. Thread topics are assigned by keyword matching over the opening post's subject and slug. 76.2% of posts fall into "Other / Misc". The topic covariate in Table 4 should be read as a coarse control, not as a substantive topical analysis.

Scope. One board, 24 daily captures, 45 days. Results describe the surviving, publicly visible discourse of /pol/ during that window. They do not generalise to /pol/ across time, nor to other imageboards.

7 Conclusion

/pol/ is not primarily arguing about the news. It is circulating platform media and archived artefacts, with legacy and alternative outlets together occupying under five percent of its citation stream and a handful of domains dominating the rest. Where tone does differ by source category, the differences are statistically clear but substantively small, and the one contrast the field's framing most confidently predicts—mainstream against alternative—does not reach significance at the sample size the board itself affords.

The productive move is not more data under the same frame. It is to take platform and archive infrastructure as the unit of analysis, to anchor source coding to external rating databases across all three axes, and to replace whole-post sentiment with target-dependent stance detection. All three are tractable; none of them are what the current framing asks for.

Data and code availability

Analysis code, derived statistics, and figures are available at `https://github.com/marcgarnier/4chan-pol-source-audit`. The raw capture corpus (~600 MB) is not redistributed; it is reproducible from the public 4chan JSON API using the collection scripts provided. All results in this paper are regenerated by `analysis.py` from the released database schema.

References

Barbieri, F., Camacho-Collados, J., Neves, L., & Espinosa-Anke, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. *Findings of EMNLP 2020.*

Bozarth, L., Saraf, A., & Budak, C. (2020). Higher Ground? How Groundtruth Labeling Impacts Our Understanding of Fake News about the 2016 U.S. Presidential Nominees. *ICWSM 2020.*

Cliff, N. (1993). Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin*, 114(3), 494–509.

Hine, G. E., Onaolapo, J., De Cristofaro, E., Kourtellis, N., Leontiadis, I., Samaras, R., Stringhini, G., & Blackburn, J. (2017). Kek, Cucks, and God Emperor Machine Learning: A Measurement Study of 4chan's Politically Incorrect Board. *ICWSM 2017.*

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621.

Loureiro, D., Barbieri, F., Neves, L., Espinosa-Anke, L., & Camacho-Collados, J. (2022). TimeLMs: Diachronic Language Models from Twitter. *ACL 2022, System Demonstrations.*

Mohammad, S., Kiritchenko, S., Sobhani, P., Zhu, X., & Cherry, C. (2016). SemEval-2016 Task 6: Detecting Stance in Tweets. *SemEval 2016.*

Papasavva, A., Zannettou, S., De Cristofaro, E., Stringhini, G., & Blackburn, J. (2020). Raiders of the Lost Kek: 3.5 Years of Augmented 4chan Posts from the Politically Incorrect Board. *ICWSM 2020.*

Pontiki, M., Galanis, D., Papageorgiou, H., et al. (2016). SemEval-2016 Task 5: Aspect-Based Sentiment Analysis. *SemEval 2016.*

Robertson, R. E., Jiang, S., Joseph, K., Friedland, L., Lazer, D., & Wilson, C. (2018). Auditing Partisan Audience Bias within Google Search. *CSCW 2018.*

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212.